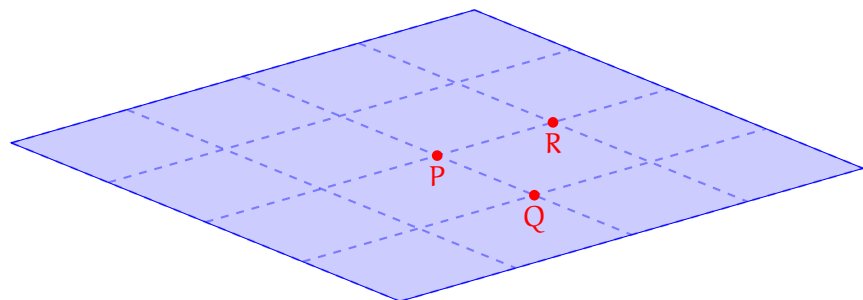


Example 1. Find a scalar equation for the plane that passes through the points $P(1, -2, 1)$, $Q(-4, 5, -8)$, and $R(7, -8, 7)$.



Solution. First we select any point on the plane:

Next we construct a normal vector the the plane using the:

Next we set up the scalar equation $\vec{n} \cdot \vec{x} = \vec{n} \cdot \vec{p}$ of the plane.

Example 2. Consider the planes:

$$\mathcal{P}_1 : 2x + y + 4z = 2$$

$$\mathcal{P}_2 : 2x - 4y + z = 2$$

(a) Find the angle between the planes.

Solution. The angle between the planes equals the angle between their:

We find them by reading of the coefficients of the variables on the lefthand side.

$$\vec{n}_1 =$$

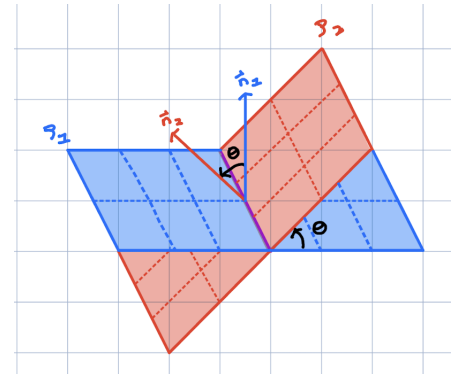
$$\vec{n}_2 =$$

Next we compute the two equivalent expressions of the dot product:

$$\vec{n}_1 \cdot \vec{n}_2 =$$

$$\|\vec{n}_1\| \|\vec{n}_2\| \cos \theta =$$

and then equating and solving for θ .



(b) Parametrize the line of intersection of the two planes.

Solution. We first identify at least one point common to both planes.

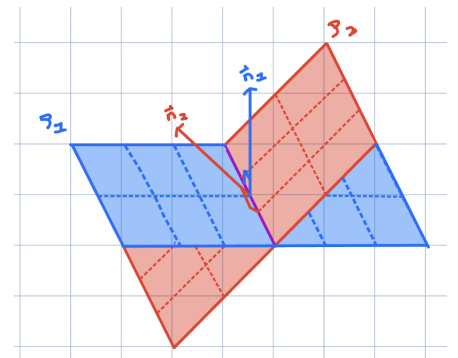
We could do this e.g. by setting:

This gives us the point: $P =$

Because the line is on both planes, it is orthogonal to both normals.

Hence, a direction vector for the line is:

Thus a parametrization for the line is:

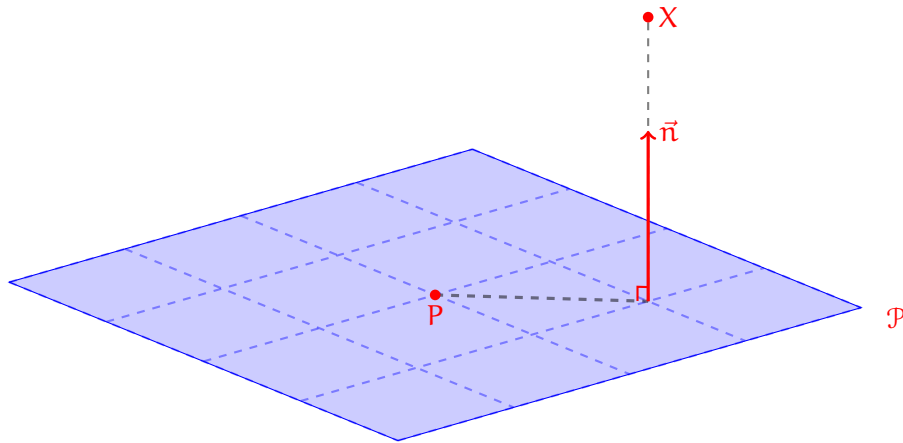


A. **Distances.** We next compute distances between points/lines/planes.

For us distance will always mean:

Let us begin with the distance between a point X and a plane $\mathcal{P}$.

We will need to know a point P on the plane and a normal $\vec{n}$ to the plane.



We sketch the distance as part of a right triangle.

This value is the component of $\overrightarrow{PX}$ that is:

In other words, the distance is:

Generally, we want distances to be nonnegative, hence we add an absolute value.

Let $\mathcal{P}$ a plane through point P with normal $\vec{n}$. If X is another point:

$\text{distance}(X, \mathcal{P}) =$

This exemplifies a recurrent theme.

To calculate distance, take the component of the displacement parallel to:

Example 3. Consider the planes:

$$\mathcal{P}_1 : 10x + 2y - 2z = 6$$

$$\mathcal{P}_2 : 5x + y - z = 1$$

(a) Explain why these planes are parallel.

Solution. Two planes are parallel if and only if their normal vectors are parallel.

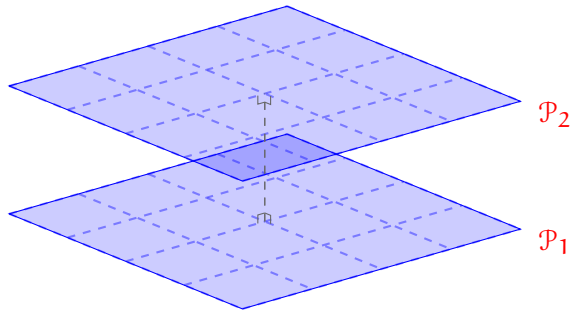
The normals are:

$$\vec{n}_1 =$$

$$\vec{n}_2 =$$

We observe that:

(b) Find the distance between the planes.



Solution. A common normal is $\vec{n} =$

We select any pair of points, one on each plane, and calculate the displacement.

The underlying principle is that the distance between two objects equals the component of a displacement vector between the two objects onto their normal vector.

We lastly compute the component using the formula $\text{comp}_{\vec{x}}(\vec{y}) = (\vec{x} \cdot \vec{y}) / \|\vec{x}\|$.

$$\text{distance}(\mathcal{P}_1, \mathcal{P}_2) =$$

(c) Find the closest point on plane $\mathcal{P}_2$ to the point P_1 from part (b).

Solution. The orthogonal line from point P_1 to plane $\mathcal{P}_2$ is parametrized by:

$$\vec{r}(t) =$$

Its intersection with the $\mathcal{P}_2$ is the closest point on the plane to P_1 .

We find it by subbing the parametrization into the eqn $5x + y - z = 1$ of $\mathcal{P}_2$.